

Multiple Choice (10 marks)

1. Explain three ways in which a person's basal metabolic rate maybe determined.
 - (a) BMR can be measured by the duration and quality of sleep
 - (b) BMR can be estimated by the level of physical activity in a daily routine
 - (c) BMR can be determined by measuring heart rate
 - (d) BMR can be determined by the amount of sweat produced during physical activity
 - (e) BMR can be estimated by the frequency of urination
2. In plants, which tissue is responsible for new growth?
 - (a) Apical tissue
 - (b) Sclerenchyma tissue
 - (c) Ground tissue
 - (d) Vascular tissue
 - (e) Cork tissue
3. Why would you not expect conjugation among a group of paramecia that had descended from a single individual through repeated fission?
 - (a) conjugation results in no genetic diversity
 - (b) conjugation requires more than two organisms
 - (c) all individuals are identical genetically, and conjugation could not occur
 - (d) repeated fission in paramecia results in sterile offspring that cannot conjugate
 - (e) the group of paramecia lacks the cellular structures necessary for conjugation
4. What factors determine the characteristics of particular organisms that exist in an ecosystem.
 - (a) Temperature, precipitation, soil chemistry, and the types of decomposers present
 - (b) Only soil structure and chemistry
 - (c) Precipitation, temperature, and the presence of specific predator species
 - (d) Only temperature and precipitation
 - (e) Only the concentration of trace elements
5. Which of the following are traits that are affected by more than one gene?
 - (a) Heterozygous traits
 - (b) Mendelian traits

- (c) Multiple alleles traits
- (d) Polygenic traits
- (e) Pleiotropic traits

True / False (10 marks)

Answer True or False

6. True or False: Do hospitals provide lower quality care on weekends?
7. True or False: Is ganglionated plexi ablation during Maze IV procedure beneficial for postoperative long-term stable sinus rhythm?
8. True or False: Does pediatric housestaff experience influence tests ordered for infants in the neonatal intensive care unit?
9. True or False: Is ankle/arm pressure predictive for cardiovascular mortality in older patients living in nursing homes?
10. True or False: Endovenous laser ablation in the treatment of small saphenous varicose veins: does site of access influence early outcomes?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the physiological and cellular implications of removing the macronucleus from a paramecium, focusing on how this alteration affects its overall viability and function.
12. Discuss the concept of heritability as it applies to the average number of abdominal bristles in *Drosophila melanogaster*, using the provided population means to calculate and analyze the heritability of this trait.

End of Paper